Lecture 2

- We saw that the general solution of a linear ODE can be expressed in terms of complementary and particular solutions.
  - Complementary solutions form a solution space.
  - For 1st-order ODEs the solution space is 1D and for 2nd-order ODEs they are 2D ...more on this today
- We learnt how to use reduction of order to solve a 2nd-order ODE when one complementary solution is known

Existence and uniqueness of solutions

Theorem

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t).$$

If f_0, f_1 and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is a linear sum of 2 linearly independent complementary solutions and a single particular solution.
  - Solutions space (i.e., all complementary solutions) is a 2-dimensional space.
- IVPs: When supplemented with initial values $y(t_0)$ and $y'(t_0)$ where $t_0 \in D$, it has a unique solution which is twice differentiable.

Linear independence

Linear algebra:

Two vectors, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, are linearly dependent if there exists c such that $\boldsymbol{v}_1 = c\,\boldsymbol{v}_2$.

Differential equations:

Two functions, $y_1(t)$ and $y_2(t)$, linearly dependent if there exists c such that

$$y_1(t) = c\,y_2(t) \quad \text{for all} \quad t.$$

Exercise 1.6: Are $y_1 = \mathrm{e}^{-2t}$ and $y_2 = 2\mathrm{e}^{-2t}$ linearly independent.

The Wronskian

Definition (The Wronskian)

Given two functions y_1 and y_2, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2](t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

- y_1 and y_2 are linearly independent if

$$\mathrm{wr}[y_1, y_2](t) \neq 0 \quad \text{for some } t$$

  - y_1 and y_2 are linearly dependent if $\mathrm{wr}[y_1, y_2](t) = 0$ for all t
  - $\mathrm{wr}[y_1, y_2] = 0$ alone tells you nothing

Theorem (Abel's differential equation identity)

Let y_1 and y_2 be two solutions of $y'' + f_1 y' + f_0 y = 0$. Then

$$\mathrm{wr}[y_1, y_2](t) = \mathrm{wr}[y_1, y_2](t_0)\mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau}$$

- If the Wronskian is non-zero anywhere it will be non-zero everywhere.
- If the Wronskian is zero anywhere it will be zero everywhere.

If we know one solution, say y_1, we can use Abel's identity and Slide 4 to find the second linearly independent solution, y_2, by solving the ODE

$$y_1 y_2' - y_1' y_2 = \mathrm{wr}[y_1, y_2](t_0)\mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau};$$

this is equivalent to using reduction of order.

Exercise 1.7: Study linear independence of solutions to below ODE

$$y'' + by' + \frac{b^2}{4}y = 0 \quad \Rightarrow \quad y(t) = C_1\,\mathrm{e}^{-\frac{b}{2}t} + C_2\,t\mathrm{e}^{-\frac{b}{2}t}$$

Exercise 1.8: Find solutions of the form $\phi = r^\lambda$ for the below ODE

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}\phi}{\mathrm{d}r} - \frac{n^2}{r^2}\phi = 0, \qquad r > 0, \qquad n \in \{0, 1, 2, \ldots\}^*$$

*This is the Euler equation from, e.g., electrostatics, which we will see again in Topic 5

Higher-order (linear) ODEs

Theorem (Solutions of an nth order linear ODE/IVP)

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^n}{\mathrm{d}t^n} + f_{n-1}(t)\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)\cdot$$

If $f_0, f_1, \ldots, f_{n-1}$ and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is the sum of n linearly independent complementary solutions and a single particular solution.
  - Solution space (i.e., all complementary solutions) form an n-dimensional vector space.
- IVPs: When supplemented with initial values $y(t_0), y'(t_0), \ldots, y^{(n-1)}(t_0)$ where $t_0 \in D$, it has a unique solution which is n-times differentiable.

Definition (The Wronskian)

Given n functions $y_1, y_2, \ldots, y_n$, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

- $y_1, y_2, \ldots, y_n$ are linearly independent if there exists a t such that

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) \neq 0$$

  - $y_1, y_2, \ldots, y_n$ are linearly dependent if $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = 0$ for all t

Method of trial solutions (higher order)

Exercise 1.9: Find the general solution to the below ODE by trialing $y = \mathrm{e}^{\lambda t}$.

$$y''' + y'' - 4y' - 4y = 0$$

Reduction of order also works in higher dimensions but we won't cover that in MAST20030.

Summary of Linear ODEs

1. An nth order linear ODE has n *linearly independent* complementary solutions and a single particular solution. It has a unique solution if n initial values are specified.

2. Check for linear independence using the Wronskian.

3. If one complementary solution for a 2nd order ODE is known, the general solution can be found through reduction of order.